

Data-Driven Generation of Synthetic Timber for Deep-Learning-Based Wood Assessment

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Data-Driven Generation of Synthetic Timber for Deep-Learning-Based Wood Assessment

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Abstract

This paper presents a framework for procedural generation of synthetic Norway spruce boards with realistic surface appearance and known internal structure. The framework is intended to support the development of data-driven methods for wood assessment tasks where real paired data are difficult to obtain. The proposed framework combines a data-driven knot sequence generator, a growth-layer model, a fiber-orientation model, and a conditional diffusion model for realistic surface-image generation. Knot sequences are generated using an LSTM model trained on knots extracted from X-ray CT scans of Norway spruce logs. Growth layers are generated using a stochastic model and are modified to account for taper, crook, and knot-induced distortions. A fiber-orientation model is then used to assign 3D fiber directions based on the generated knot and growth-layer geometry. Finally, a modified Stable Diffusion XL model translates the generated surface ring patterns and fiber-orientation maps into realistic RGB images of the board faces. The study also presents a preliminary application of this framework, in which synthetic boards were used to train a model for predicting the cross-sectional ring pattern of a timber board from images of its four longitudinal surfaces. The board generator is made openly available at https://github.com/osamaoa/board_generator.

1. Introduction

Wood is a naturally grown material with substantial variation in appearance, anatomy, and physical and mechanical properties. The stiffness and strength within a timber element vary locally depending on wood features such as growth patterns, knot size and type, and the fiber deviations associated with knots (Isaksson 1999; Dinwoodie 2002). These features are also important for the shape stability of timber, since moisture-related distortions such as cupping, twisting, and bowing depend on the local material directions within the timber element (Ormarsson et al. 2000). The visual quality of timber products is also affected by features such as knots, pith, annual rings, blue stain, wane, and cracks (Urbonas et al. 2019). Due to this high level of natural variation, optimal use of wood requires accurate grading with respect to both appearance and mechanical performance.

Modern sawmills increasingly use high-resolution multi-sensor scanning systems to obtain detailed information about timber boards. Optical cameras, laser scanning based on the tracheid effect, X-ray scanning, and other non-destructive measurement techniques can provide information about surface appearance, fiber orientation, density variations, knots, and other features relevant to grading. Industrial laser scanners can measure in-plane fiber directions on timber surfaces, and such information has been shown to be useful for assessing local stiffness and strength variations in sawn timber (Kandler et al. 2016; Besseau et al. 2020). These developments have made it possible to move beyond grading methods based mainly on global indicators, such as dynamic modulus of elasticity, toward methods that use local information about material structure and defects (Briggert et al. 2020).

However, optical and laser scanning of timber boards produce large amounts of data that must be processed before they can be used in grading and assessment. Extracting relevant information from such data is challenging because of the significant variation in the appearance of wood features. The task is made more difficult by the high speed requirements of industrial production lines, where scanning speeds often exceed 200 m/min (Habite et al. 2020). For a feature-extraction method to be useful in industrial grading, it must therefore be accurate, robust, and computationally efficient. In response to these requirements, wood-scanning research has increasingly moved away from conventional image-processing pipelines, which are based on user-defined thresholds and hand-crafted filters, toward data-driven deep learning (DL) models that learn relevant features directly from examples.

Several recent studies have demonstrated the potential of object-detection DL models for detecting wood features and defects that are directly visible on timber or wood-product surfaces. For example, Urbonas et al. (Urbonas et al. 2019) used a Faster R-CNN with transfer learning and data augmentation to detect and classify defects on wood veneer surfaces, including knots and cracks. Ding et al. (Ding et al. 2020) proposed an improved model based on a Single Shot MultiBox Detector (SSD) for detecting multiple types of defects on solid wood panels, including live knots, dead knots, and cracks. Fang et al. (2021) applied YOLOv5 to detect knots on sawn timber surface. Beyond object detection, pixel-level image segmentation has also been investigated. For instance, Zhu et al. (2024) proposed a U-Net-based network to segment wood surface defects, such as cracks and dead knots, by combining RGB image data with depth information from laser profile sensors. In these studies, the ground truth is typically obtained by manually annotating the relevant features directly on the scanned surfaces.

This type of supervised learning faces two important challenges. First, DL models generally require large and diverse datasets for training and validation. Manual annotation of wood features is laborious and time-consuming, especially when pixel-level segmentation is needed. As a result, many studies use most of the available annotated data for training and validation, leaving relatively limited data for independent testing. This can make it difficult to assess how well the trained models generalize to new boards, production lines, scanners, or species. Second, and more importantly, for many practical applications, the target information is not directly visible on the board surfaces. In such cases, the ground truth cannot be obtained by simple manual annotation of optical images. Instead, special experimental methods are needed to determine the internal structure of the timber, including pith location, annual ring pattern, knot geometry, and three-dimensional fiber orientation.

One example is the prediction of pith location and annual ring patterns at arbitrary positions along a board from external surface scans. This information is relevant because pith location and annual ring width are related to material properties, shape stability, and visual grading (Blouin et al. 1994; EN 1611–1:1999 1999; Ormarsson et al. 2000). However, the ground truth along full boards can only be obtained by physically cutting the boards at multiple positions or by using X-ray computed tomography (CT) scanning (Longuetaud et al. 2004; Huber et al. 2022). Another example is the prediction of three-dimensional knot geometry from surface information. Surface scanning can detect visible knot areas and provide information about fiber deviations on the surfaces. However, acquiring accurate ground truth for the full internal knot geometry at scale would require X-ray CT scanning of a large number of boards or logs, which is expensive and time-consuming (Hu et al. 2025). A third example is the prediction of full three-dimensional fiber orientation from surface images and in-plane fiber directions obtained by wood scanners. Current methods for measuring three-dimensional fiber orientations are destructive and time-consuming, as they involve progressive planing and scanning to expose the fiber orientation on internal surfaces (Penvern et al. 2024; Hu et al. 2025). While such methods are valuable for model development and validation, they are not practical for producing the large paired datasets required to train DL models.

When it is difficult or impossible to obtain sufficiently large real paired datasets, synthetic data offer a potential solution. This idea has been explored extensively in applications such as autonomous driving, medical imaging, and other fields where reliable annotations are costly or time-consuming to produce. In autonomous driving, synthetic datasets such as SYNTHIA have been generated from virtual environments with automatic pixel-level semantic labels, avoiding the need for manual annotation of every object boundary in real images (Ros et al. 2016). Similarly, images extracted from commercial computer games have been used to generate pixel-accurate semantic labels for driving scenes, showing that synthetic game data can improve semantic segmentation performance and reduce the amount of manually labeled real data needed for training (Richter et al. 2016). Similar solutions have been proposed in medical imaging, where synthetic images have been used to address the rarity of certain conditions, the high cost of expert annotation, and ethical and privacy concerns related to the use of patient data. For example, Frid-Adar et al. (Frid-Adar et al. 2018) used generative adversarial networks to synthesize CT images of liver lesions and showed that adding synthetic lesions to the training data improved CNN-based lesion classification.

The use of synthetic data in wood assessment has already shown promising results. Habite et al. (Habite et al. 2022) developed a stochastic model to generate virtual Norway spruce boards with known pith location and realistic surfaces, and used these boards to train a one-dimensional convolutional neural network for pith detection from greyscale images of the four longitudinal board sides. The trained model was successfully applied to real boards, demonstrating that synthetic boards can provide useful supervision for a task where large real datasets with known pith location are difficult to obtain. However, the scope of that generator was limited to knot-free clear wood sections of Norway spruce boards, and it did not model knots or three-dimensional fiber orientation. Therefore, its possible applications were mainly limited to pith and annual-ring inference in clear wood.

Another study relevant to synthetic wood modeling is the procedural framework proposed by Larsson et al. (2022) for modeling the annual ring pattern of solid wood with knots. Their method takes the skeletal structure of a tree log as input and produces a three-dimensional scalar field representing the relative time at which different parts of the wood were formed. This field is used to define the volumetric annual ring pattern. Separate fields are first computed for the stem and for each knot, and these are then merged to reproduce the transition between stem wood and knots. This work is an important contribution to procedural wood texturing, especially for visual rendering. However, it focuses on generating plausible volumetric wood textures from prescribed tree and knot skeleton data, rather than on generating large datasets of complete sawn timber boards with stochastic knot sequences, growth-layer geometry, and scanner-like surface images. In addition, the framework does not calculate three-dimensional fiber orientations within the generated timber elements.

The aim of the present study is to develop a more complete framework for procedural generation of synthetic Norway spruce boards. Unlike previous wood-generation approaches (Habite et al. 2022; Larsson et al. 2022), the proposed framework generates boards with realistic surface appearance and known internal structure, including knot geometry, growth-layer geometry, and three-dimensional fiber orientation. The purpose is to support future studies on DL-based characterization of sawn timber by providing paired synthetic data for tasks where real ground truth is difficult, costly, or destructive to obtain.

The detailed objectives of the study are as follows:

1. To develop a data-driven knot generator capable of generating realistic knot sequences similar to those observed in real Norway spruce trees.
2. To develop a growth-layer and fiber-orientation model capable of handling multiple knots and irregular growth layers with crook and taper.
3. To train a DL model that translates generated ring patterns and fiber orientations into realistic surface images of Norway spruce.

As a proof of concept, we also present a preliminary application in which synthetic boards generated by the proposed framework were used to train a model for predicting timber cross-sections from the four external board surfaces. This application illustrates how the generator can provide paired training data for a task where large-scale real ground truth would be difficult to obtain, and gives an initial indication of how models trained on generated boards can transfer to real timber.

2. Method overview

The proposed framework consists of four connected components that together generate synthetic Norway spruce boards with known internal structure and realistic external appearance. The overall workflow is illustrated in Figure 1. First, a knot generator is used to create a discrete sequence of knots along a virtual log. Second, a growth-layer generator is used to create annual-ring geometry, including irregular ring shapes, variable ring widths, taper, crook, and local distortions around knots. Third, a fiber orientation model assigns a three-dimensional fiber direction to each point in the generated board based on the knot geometry and the surrounding growth layers. Finally, a diffusion-based image-generation model produces realistic RGB images of the board faces.

After the internal structure has been generated, the board is virtually sawn into the desired dimensions. The intersections between the growth layers and the board surfaces provide the visible ring patterns, while the three-dimensional fiber-orientation field is projected onto the surfaces to obtain in-plane fiber-orientation maps. These surface maps are then used as conditioning information for a diffusion-based image-generation model, which produces realistic RGB images of the four board faces. The result is a synthetic board for which both the external surface appearance and the underlying internal structure are known. The following sections describe each part of the framework in more detail.

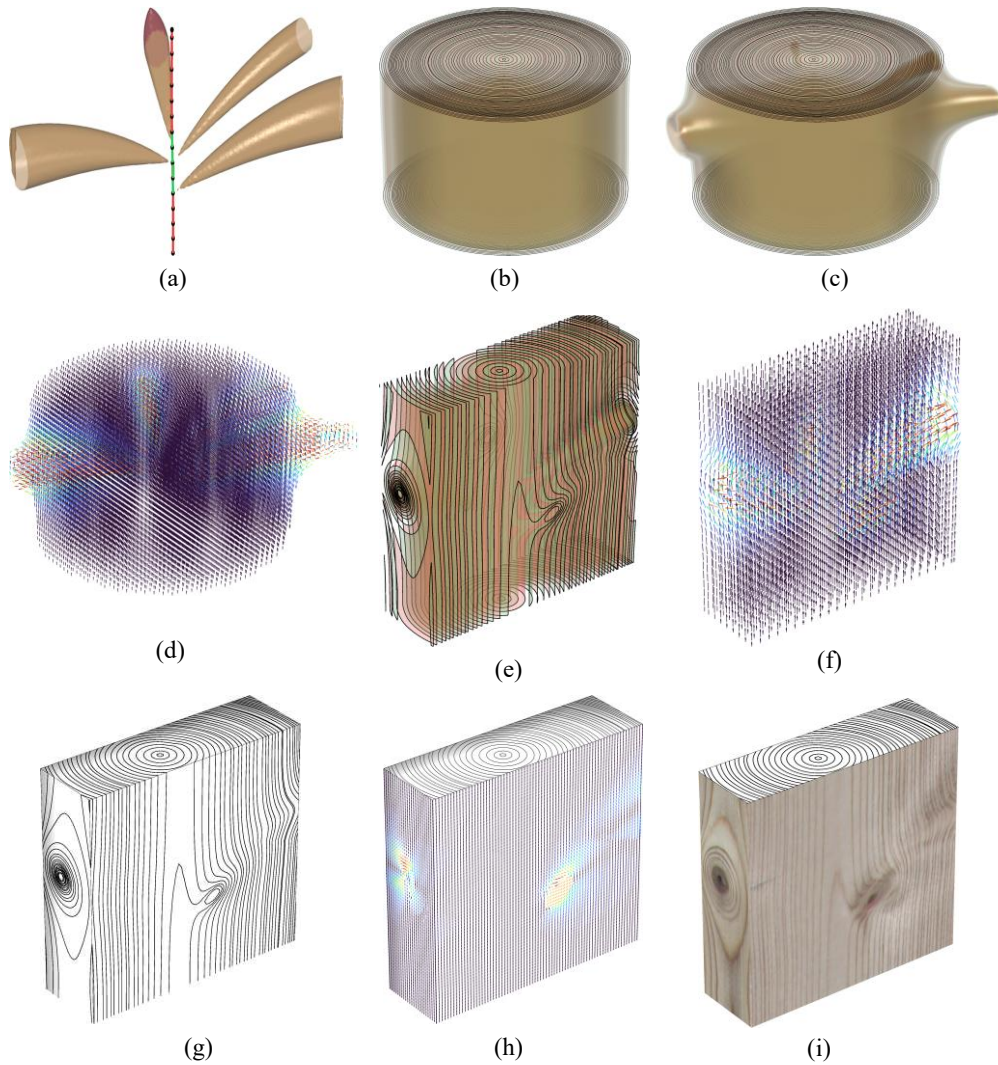


Figure 1. Steps for generating synthetic boards using the proposed framework: (a) The knot generator generates a knot sequence. (b) Growth layers are generated using the growth layer model. (c) The growth layers are modified to account for knot bumps. (d) 3D fiber orientations are calculated with the generated log. (e) A board is virtually sawn from the log. (f) 3D fiber orientations within the board are obtained. (g) Binary ring maps are obtained for the four faces. (h) In-plane fiber orientations maps are obtained for the four faces. (i) A diffusion model is used to translate the ring and fiber maps to realistic timber surfaces.

3. Knot generator

The knot generator is a long short-term memory (LSTM) model trained to produce realistic knot sequences, resembling those observed in Norway spruce trees. Much like autoregressive language models used in text generation, it learns to predict the next token in a sequence that encodes knot occurrences along a log. Each token corresponds to a discrete position and indicates either the absence of a knot or the presence of a specific knot, sampled from a comprehensive dictionary derived from real Norway spruce logs. Figure 2 illustrates the overall concept of the knot generator. Further details on dataset preparation, model training, and validation are provided in the following subsections.

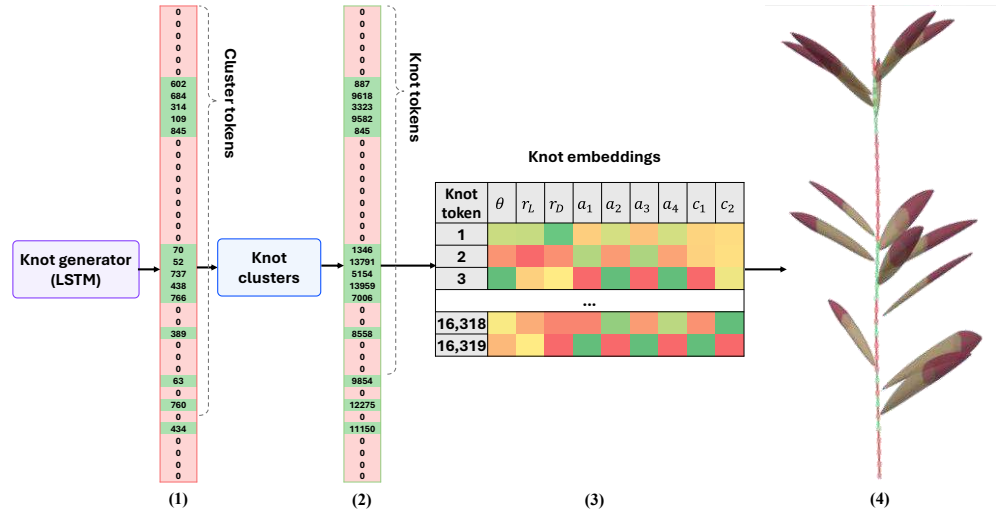


Figure 2. Schematic of the knot generation procedure: 1- The LSTM model generates a sequence of cluster tokens. 2- Knots are randomly sampled from each corresponding, resulting in a sequence of knot tokens. 3- The nine knot geometry parameters are obtained for each knot according to the knot embedding matrix. 4- The knot sequence is rendered.

3.1 Dataset and knot parameterization

The training data were extracted from the open-access database TreeTrace_spruce (Longuetaud et al. 2023), which contains volumetric X-ray CT images of 97 Norway spruce logs, each 4 m long. The image resolution was 5 mm in the longitudinal direction and 1.1 mm in the transverse direction. The CT images in the dataset were segmented such that different wood features, including the pith, live knots, dead knots, bark, sapwood, and heartwood, are represented by distinct intensity values. According to information provided by the dataset's authors, this segmentation was carried out using the CTPro software package (Zhou et al. 2025). Based on these segmented volumes, point clouds were extracted for the pith and for the live and dead knot regions. In a small number of cases, neighboring knots were still connected in the segmented data and were therefore separated in Avizo software (Thermo Fisher Scientific, Waltham, MA, USA) using a 3D morphological opening operation.

After this preprocessing, each knot was represented by a separate point cloud, which was then fitted using the knot model proposed by Foley (2003). According to this model, the geometry of a knot, i , can be described in terms of the following 10 parameters, which are illustrated in Figure 3:

- The longitudinal starting position of the knot along the pith of the log $z_{0,i}$.
- The azimuth of the knot in the (x, y) plane, $\theta_{0,i}$.
- The coefficients of the knot axis parabola, $c_{1,i}$ and $c_{2,i}$.
- The coefficients of the quartic live knot diameter expression, $a_{1,i}, \dots, a_{4,i}$.
- The distance in the radial direction from the pith to the boundary between the live and dead parts of the knot, $r_{L,i}$.
- The distance in the radial direction from the pith to the end of the knot, $r_{D,i}$.

As shown in Figure 3, the knot axis $Z_{c,i}$ and diameter L_i are defined in the local (ξ_i, z) plane as:

$$Z_{c,i}(\xi_i) = c_{1,i}\xi_i^2 + c_{2,i}\xi_i + z_{0,i} \quad (1)$$

$$L_i(\xi_i) = a_{1,i}\xi_i^4 + a_{2,i}\xi_i^3 + a_{3,i}\xi_i^2 + a_{4,i}\xi_i \quad (2)$$

As a result of this knot extraction and parameterization process, a total of $N_k = 16,319$ parameterized knots were obtained from the 97 logs. Each log is now represented by an array with 10 columns, corresponding to the 10 knot parameters.

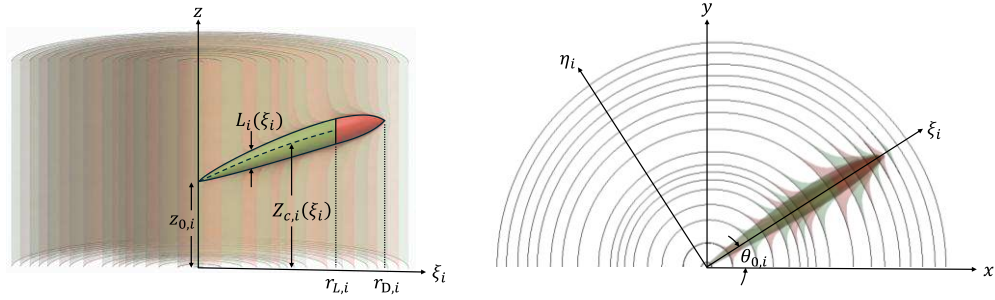


Figure 3. Description of the knot geometry parameters. The coordinate ξ_i denotes the local direction along the i^{th} knot axis, measured outward from the pith, while η_i denotes the local direction perpendicular to ξ_i in the cross-sectional plane.

3.2 Log discretization

The parametrized representations of the knots for each log were used to generate training sequences for the LSTM model. To do so, all knots from the 97 logs were given a unique knot token from 1 to 16,319. Each log was then discretized into a long sequence of 1-cm slots, yielding 400 slots per log. Each slot was assigned 0 if it does not contain a knot, or the corresponding knot token if it contains one. If two knots landed in the same slot, one of them was shifted forward by one slot.

Each log then becomes a discrete sequence where 0 denotes an empty slot, and every nonzero token refers to one knot dictionary entry. After discretization, each knot can be described by 9 rather than 10 parameters, because its longitudinal starting position $z_{0,i}$ is implicitly determined by the knot's position in the sequence. Each log sequence is segmented into overlapping sequences with a length of 50 tokens and a stride of 1 with no padding, resulting in a total of 37,680 sequences.

3.3 Knot clustering and embedding

In contrast to words in natural language, which may appear many times in the training data in many different sequences, an individual knot appears only once in a full log sequence. After the log has been divided into overlapping subsequences, that same knot is repeated only a few times in the training data. Therefore, if every knot is assigned its own unique token, the resulting “vocabulary” becomes large, while most knot tokens occur only rarely. This gives a sparse representation of the data and makes it difficult for the LSTM model to learn reliable sequential patterns.

Another limitation of using unique knot tokens is that they treat all knots as completely distinct, even when many of them have very similar geometric properties. In practice, knots from different logs may still share similar shapes, orientations, and radial extents. If similar knots are represented by different tokens, the model cannot easily exploit these similarities.

To address this issue, the knots were grouped into clusters in the space of knot parameters. In this way, knots with similar geometry are represented by the same token rather than by separate unique identifiers. This reduces the size of the vocabulary and increases the number of times a given token appears across the training data. As a result, the model is provided with more repeated examples of similar knot types, which makes the learning problem more statistically efficient.

Each knot is represented by the following 9-dimensional embedding vector:

$$\mathbf{k}_i = (\theta_{0,i}, c_{1,i}, c_{2,i}, a_{1,i}, a_{2,i}, a_{3,i}, a_{4,i}, r_{L,i}, r_{D,i}) \quad (3)$$

Before clustering, each component of k_i was normalized to the interval $[0,1]$ across the dataset. The normalized knot vector $\hat{\mathbf{k}}_i$ is defined component-wise as:

$$\hat{k}_{ij} = \frac{k_{ij} - \min_m(k_{mj})}{\max_m(k_{mj}) - \min_m(k_{mj})}, \quad j = 1, \dots, 9 \quad (4)$$

where k_{ij} denotes the j -th component of the vector for knot i , and the minimum and maximum are taken over all knots in the dataset. The normalized vectors were grouped into Q clusters using the k-means method. The clustering problem is formulated as:

$$\min_{\{\mu_q\}, \{c_i\}} \sum_{i=1}^{N_k} \|\hat{\mathbf{k}}_i - \mu_{c_i}\|_2^2, \quad c_i \in \{1, \dots, Q\} \quad (5)$$

where μ_q is the centroid of cluster q and c_i is the assigned cluster index of i^{th} knot vector.

After clustering, each cluster is represented by a unique token from 1 to Q , and the corresponding centroid μ_q is used as an embedding for that cluster. The zeroth token is reserved for the no-knot state, which is represented by a zero vector. This results in an embedding matrix with $Q + 1$ rows and 9 columns. The knot tokens in the overlapping sequences, described in Section 3.2, are replaced by their corresponding cluster tokens. In this work, the number of clusters was chosen as $Q = 1000$ as a compromise between detail and repetition in the training data. Using more clusters would preserve finer differences between knots, but would also make each cluster token less common, whereas fewer clusters would place a wider range of knots in the same group and reduce the level of detail available to the model.

3.4 LSTM model

The knot sequence model was trained using the overlapping token sequences described in Sections 3.2 and 3.3. Let $\mathbf{x}$ denote one training sequence of length t . The sequence can be written as:

$$\mathbf{x} = (x_1, \dots, x_{t-1}, x_t) \quad (6)$$

During training, the first $t - 1$ tokens were used as input and the final token in the sequence was used as the target. In this work, the sequence length was taken as $t = 50$, so each training example consisted of an input sequence of 49 tokens and one target token.

The model consists of a fixed embedding layer followed by two LSTM layers and then an output layer with softmax activation. The embedding layer was assigned with the $(Q + 1) \times 9$ embedding matrix defined in Section 3.3 and was kept fixed during training. Denoting the embedding matrix by $\mathbf{E}$, the embedding vector corresponding to the token at position n is written as:

$$\mathbf{e}_n = \mathbf{E}[x_n], \quad n = 1, \dots, t - 1 \quad (7)$$

The output of the embedding layer (i.e., the resulting sequence of embedding vectors) is then processed by two LSTM layers, each with a hidden size of 512. Denoting the hidden state and memory state at position n by $\mathbf{h}_n$ and $\mathbf{m}_n$, respectively, the recurrent update is written as:

$$(\mathbf{h}_n, \mathbf{m}_n) = \text{LSTM}(\mathbf{e}_n, \mathbf{h}_{n-1}, \mathbf{m}_{n-1}), \quad n = 1, \dots, t - 1 \quad (8)$$

The output layer maps the final hidden state to a probability distribution over the $Q + 1$ possible next tokens as follows:

$$p = \text{softmax}(\mathbf{W} \mathbf{h}_{t-1} + \mathbf{b}) \quad (9)$$

where $\mathbf{W}$ and $\mathbf{b}$ are the weights and biases of the output layer.

The model was trained by minimizing the cross-entropy loss associated with the target cluster token. A total of 33,912 sequences (90% of the sequences extracted from the dataset) were used for training, and the rest were reserved for validation. Training was conducted using PyTorch (Paszke et al. 2019) with AdamW optimizer, a learning rate of 0.001, and a batch size of 64. Early stopping was applied based on the validation loss with a patience of 4 epochs. The training was terminated after 75 epochs. The accuracy of the trained LSTM in predicting the target cluster token, evaluated over the 3768 validation sequences, was 93.2%.

3.5 Generation and statistical evaluation of knot sequences

After training, the model was used to generate random knot sequences resembling those observed in Norway spruce trees. The generation process starts from a seed sequence, which in this work consists of a single zero token representing a no-knot state at the first 1 cm position of the generated log. This seed sequence is fed to the model to obtain a probability distribution over the $Q + 1$ possible next cluster tokens. The next token is then selected using top- p sampling. In this method, the candidate set consists of the most probable tokens whose cumulative probability reaches a prescribed threshold (in this work a threshold of 0.8 was used). The sampled token is appended to the sequence and fed back to the model, and this procedure is repeated until the desired sequence length is reached. The resulting cluster sequence is finally translated into a knot sequence by sampling one knot at random from the corresponding knot cluster for each nonzero token.

To assess the realism of the generated sequences, the trained model was used to generate 1000 sequences of length 50, and these were compared with the 37,680 sequences extracted from the CT data. Overall, the generated sequences reproduced the main statistical properties of the training sequences well. The proportion of segments containing knots was 31.81% in the generated data, compared with 34.69% in the training data, while the average number of knots per sequence was 15.903 compared with 17.343. The average knot-run length (i.e., the average number of consecutive slots containing knots) was also very similar, with values of 1.985 and 2.026 slots for the generated and training sequences, respectively. Likewise, the average continuous no-knot run length was 3.923 slots in the generated data and 3.684 slots in the training data. This comparison is presented in Table 1.

A comparison of distributions of the nine knot parameters between the generated and training data also showed good agreement. As shown in Figure 4, for all examined parameters, the means, standard deviations, and selected quantiles of the generated distributions were close to those of the training sequences. These results indicate that the proposed knot generation procedure is able to reproduce both the sequential arrangement of knots and the overall distribution of knot geometries with good fidelity.

Table 1. Comparison between actual knot sequences and those generated using the knot generator.

Metric	Training	Generated	Difference
Segments with knots	34.69%	31.81%	-2.88%
Average knot-run length (slots)	2.026	1.985	-0.041
Average continuous no-knot length (slots)	3.684	3.923	+0.239

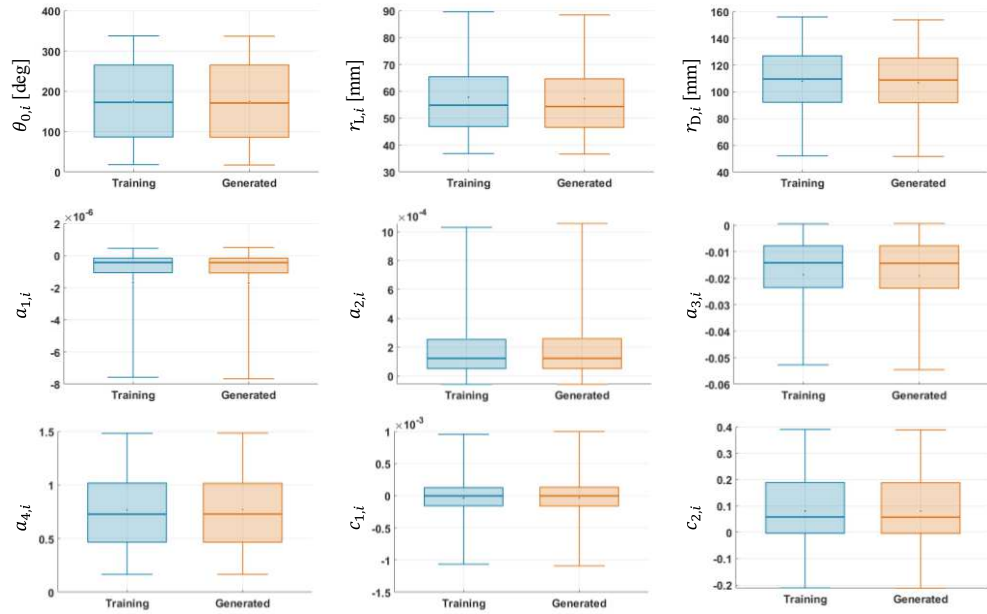


Figure 4. Comparison between the distributions of the nine knot parameters over the actual and generated knot sequences.

4. Growth layer generator

The geometry of annual rings in Norway spruce logs is influenced by factors such as natural growth, climate, tree genetics, competition from nearby trees, and damage caused by various external events (Habite et al. 2022). As a result, logs do not have perfectly cylindrical shapes. In general, trees taper upward and often exhibit some degree of crook. In addition, the cross-section of a tree or log may deviate considerably from a circular shape, and the spacing between annual rings is not uniform. Annual rings are typically wider near the pith and become narrower toward the bark in older trees. Therefore, a realistic growth layer generator should account for irregular ring shapes, variations in ring width, taper, and crook along the log. Another important phenomenon that should be considered is the local distortion of growth layers around knots, often referred to as knot bumps.

A nominal growth layer with irregular cross-sectional shape and taper, but without crook or knot bumps, can be expressed by the following scalar field:

$$g_n^{\text{nom}}(x, y, z) = x^2 + y^2 - \tilde{R}_{o,n}(\Theta, z)^2 \quad (10)$$

$$\tilde{R}_{o,n}(\Theta, z) = R_{o,n}(\Theta)(1 - t(z)) > 0 \quad (11)$$

The nominal growth-layer surface (Figure 5a) is given by the zero-level set $g_n^{\text{nom}}(x, y, z) = 0$. The coordinates x , y , and z define a Cartesian coordinate system with the origin located at the pith of the root end of the uncrooked log, Θ denotes the angular position in the cross-section, $R_{o,n}(\Theta)$ describes the cross-sectional shape of growth layer n as a function of Θ , and $t(z)$ denotes the relative taper of the tree as a function of the longitudinal position z . In the present implementation, a linear taper of $t(z) = z/80$ was assumed based on a simplification of results from previous research (Sharma et al. 2016).

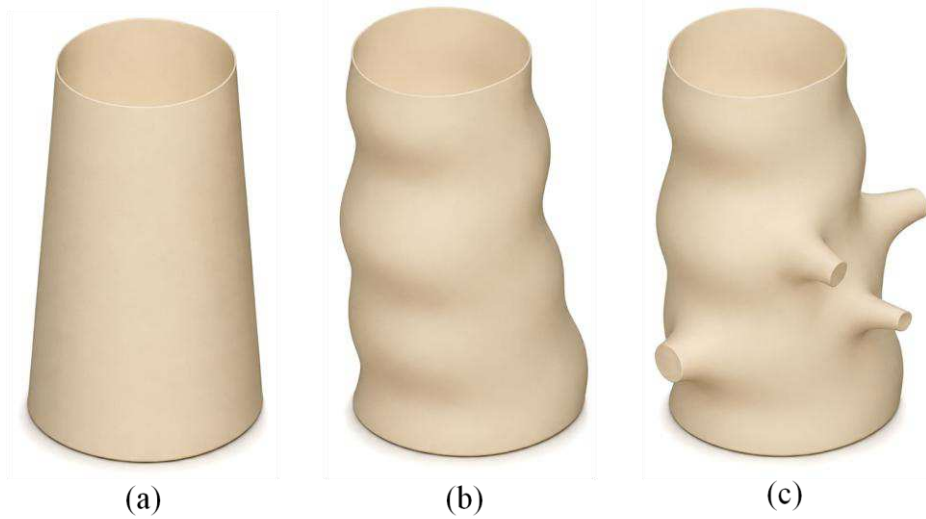


Figure 5. Illustration of growth-layer deformation. (a) Nominal, uncrooked growth layer with linear taper. (b) Growth layer after applying crook deformation. (c) Growth layer after incorporating knot-induced bumps. Note that the taper and crook are greatly exaggerated for visualization.

4.1 Growth rings shape and spacing

In this work, $R_{o,n}(\Theta)$ was obtained using a stochastic ring model previously developed by the authors (Habite et al. 2022). This model generates realistic annual-ring shapes and spacings based on measurements from real Norway spruce logs. In the stochastic model, the shape of a log cross-section is represented by a set of

correlated radii extending from the pith to the bark in different angular directions. The correlation between radii decreases with increasing angular separation, which makes it possible to generate cross-sections ranging from nearly circular to more irregular shapes. Annual ring profiles are then assigned to the generated cross-sections so that the distances between successive rings resemble those observed in real Norway spruce logs.

The model was established using data from 390 cross-sections of Norway spruce logs collected from 22 stands in Sweden and Finland. For each cross-section, radial annual-ring widths were measured from bark to bark through the pith. These data provided information both on the radial spacing of annual rings and on the overall shape of the log cross-sections. The measured log-disc data were used as the statistical basis for generating virtual log cross-sections, after which annual-ring profiles from real discs were assigned to generated shapes with similar average radius. The reader is referred to (Habite et al. 2022) for more details on the underlying data and the stochastic ring model. Figure 6 shows examples of log cross-sections generated using this model.

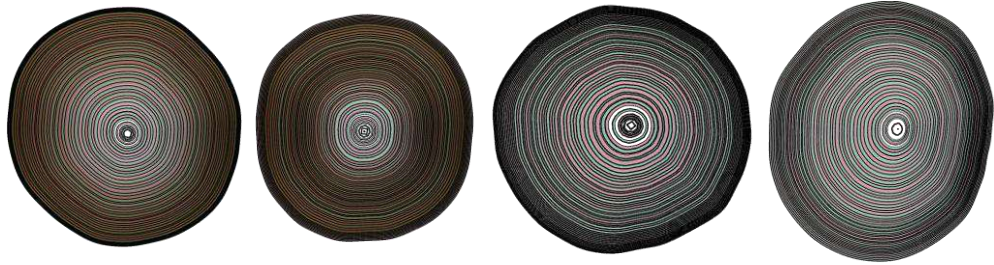


Figure 6. Example of irregular ring cross-sections generated by the ring generator.

4.2 Modeling of crook

According to the model proposed in (Habite et al. 2022), crook is represented by imposed stochastic shifts of the growth layer field in the transverse board directions (see Figure 5b). Accordingly, the transverse coordinates used in the field description are modified as functions of the longitudinal position z as follows:

$$x_c = x + D_x, \quad y_c = y + D_y \quad (12)$$

$$D_x = \sum_{j=1}^{n_c} \sin(\varphi_j) m_j \left[\frac{2\pi(z + z_{s,j})}{l_j} \right], \quad D_y = \sum_{j=1}^{n_c} \cos(\varphi_j) m_j \left[\frac{2\pi(z + z_{s,j})}{l_j} \right] \quad (13)$$

where n_c denotes the number of wavelengths considered in the crook model, l_j is the wavelength taken as $l_j = 2^{(5-j)} 1000$ [mm], $m_j \sim U(0, l_j/320)$ is a stochastic crook amplitude, $z_j \sim U(0, 4000)$ [mm] is a stochastic shift of the crook wave in the longitudinal direction, and φ_j is a random angle in radians.

The nominal scalar field in equations 10 and 11 is therefore replaced by the corresponding crooked field, given as:

$$g_n^{\text{crook}}(x, y, z) = x_c^2 + y_c^2 - \tilde{R}_{o,n}(\Theta_c, z)^2, \quad \Theta_c = \text{atan2}(y_c, x_c) \quad (14)$$

4.3 Modeling of knot bumps

The geometry of a crooked growth layer within the modeled log is further influenced by the presence of knots. After generating the knots using the knot generator and the

crooked growth layers as described above, the bump of each growth layer around the live and dead parts of the generated knots is calculated. The knot bump model developed by Foley (Foley 2003) was adopted, but with modifications to allow bump contributions from multiple knots and to account for crook and taper, rather than only a single knot bump on cylindrical growth layers as in the original model.

According to this model, the influence of the i^{th} knot on the n^{th} crooked growth layer can be determined by the following metrics:

$$p_i = \sqrt{\Delta z_i^2 + \frac{1}{k_{p,i}^2} (\xi_i^2 + \eta_i^2) \text{atan2}(\eta_i, \xi_i)^2} \quad (15)$$

$$p_{\max,n,i} = \left(\frac{A_{\text{bump},n} \tilde{R}_{o,n}^{A_{\text{exp},i}-1}}{1 - k_i} \right)^{1/B_{\text{bump},i}} \quad (16)$$

where ξ_i and η_i are the local coordinates of knot i , calculated by rotating of the crook-corrected coordinates according to the knot azimuth:

$$\xi_i = x_c \cos \theta_{0,i} - y_c \sin \theta_{0,i}, \quad \eta_i = x_c \sin \theta_{0,i} + y_c \cos \theta_{0,i} \quad (17)$$

The longitudinal offset from the knot axis Δz_i is calculated as:

$$\Delta z_i = z - Z_{c,i}(\xi_i) \quad (18)$$

The empirical parameters k_i , $k_{p,i}$, and $B_{\text{bump},i}$ were taken as 0.99, 0.95 and 2.00 for all knots, while the parameters $A_{\text{exp},i}$ and $A_{\text{bump},i}$ are defined as (Foley 2003):

$$A_{\text{exp},i} = 0.0056 L_i(\xi_i = 100) + 1.96 \quad (19)$$

$$A_{\text{bump},i} = 0.0217 L_i(\xi_i = 100) - 0.2 \quad (20)$$

The effective radius of growth layer n considering the bump around knot i is given by:

$$\hat{R}_{n,i} = \begin{cases} \tilde{R}_{o,n}, & p_i > p_{\max,n,i} \text{ or } \xi_i < 0 \text{ or } \tilde{R}_{o,n} > r_{D,i}, \\ R_{n,i}^{\text{dead}}, & p_i \leq p_{\max,n,i} \text{ and } r_{L,i} < \tilde{R}_{o,n} \leq r_{D,i} \text{ and } \xi_i \geq 0 \\ k_i \tilde{R}_{o,n} + A_{\text{bump},i} \tilde{R}_{o,n}^{A_{\text{exp},i}} p_i^{-B_{\text{bump},i}}, & p_i \leq p_{\max,n,i} \text{ and } \tilde{R}_{o,n} \leq r_{L,i} \text{ and } \xi_i \geq 0. \end{cases} \quad (21)$$

The first branch of this piecewise equation represents the growth layer outside the influence zone of the knot. The second and third branches correspond to the knot bump within the influence zone, around the dead and live parts of the knot, respectively. The radius around the dead part of the knot is expressed as:

$$R_{n,i}^{\text{dead}} = K_{\text{red},n,i} r_{D,i} + (1 - K_{\text{red},n,i}) (S_i + T_i p_{n,i,\text{alt}}^{-B_{\text{bump},i}}) \quad (22)$$

Where the associated quantities S_i , T_i , $K_{\text{red},n,i}$, and $p_{n,i,\text{alt}}$, which account for the reduction of the knot bumps, are given as:

$$S_i = k_i r_{L,i}, \quad T_{n,i} = A_{\text{bump},i} r_{L,i}^{A_{\text{exp},i}}, \quad K_{\text{red},n,i} = \frac{\tilde{R}_{o,n} - r_{L,i}}{r_{D,i} - r_{L,i}} \quad (23)$$

$$p_{n,i,\text{alt}} = p_i + \sqrt{(r_{D,i} - r_{L,i})^2 - \gamma^2 (r_{D,i} - \tilde{R}_{o,n})^2} - \sqrt{1 - \gamma^2} (r_{D,i} - r_{L,i}) \quad (24)$$

The parameter γ was taken as 2/3 (Foley 2003). A separate scalar field is then defined for each individual knot and growth layer:

$$g_{n,i}^{\text{bump}}(x, y, z) = x_c^2 + y_c^2 - \hat{R}_{n,i}^2 \quad (25)$$

All scalar fields corresponding to growth layer n are then combined by taking the minimum scalar value across all knots:

$$g_n(x, y, z) = \min_i(g_{n,i}^{\text{bump}}(x, y, z)) \quad (26)$$

This produces a single scalar field for growth layer n that includes the bump effects of all knots. The deformed growth-layer surface (Figure 5c) is finally obtained as the zero-level isosurface of the combined scalar field:

$$g_n(x, y, z) = 0 \quad (27)$$

5. Fiber orientation model

The next step in the board generator is to assign a local fiber direction to each position in the generated log. The fiber orientation model adopted here is based on the paradigm proposed by Foley (2003). However, Foley's single-knot fiber-orientation model for cylindrical growth layers was extended so that fiber directions can be calculated within logs containing multiple knots, irregular growth-ring shapes, taper, crook, and knot-induced growth-layer distortions. In that model, the fiber direction around a knot is first determined in the local (ξ_i, η_i) plane. The projected direction is then extended to 3D by forcing the fiber direction to lie within the growth layer.

Let $g_n(x, y, z)$ denote the scalar field of growth layer n , as defined in Section 4 after accounting for taper, crook, and knot bumps. At each material point, the growth layer used for the fiber-orientation calculation was taken as the nearest growth layer, defined as the layer with the smallest absolute scalar-field value at that point. The local normal direction to this growth layer is given by:

$$\mathbf{n}_n = \frac{\nabla g_n}{\|\nabla g_n\|} \quad (28)$$

The fiber direction $\mathbf{f}$ is assumed to lie within the local growth-layer surface. Therefore, it must be orthogonal to the growth-layer normal:

$$\mathbf{f} \cdot \mathbf{n}_n = 0 \quad (29)$$

This condition is used to determine the missing radial component of the fiber direction after the projected fiber direction in the local longitudinal-tangential plane around a knot has been calculated.

For each knot i , the crook-modified transverse coordinates x_c and y_c are transformed into the local knot coordinates ξ_i and η_i using equations (17) and (18). The corresponding local directions, expressed in the global coordinate system, are:

$$\mathbf{e}_{\xi,i}^b = (\cos \theta_{0,i}, -\sin \theta_{0,i}, 0), \quad \mathbf{e}_{\eta,i}^b = (\sin \theta_{0,i}, \cos \theta_{0,i}, 0), \quad \mathbf{e}_z = (0, 0, 1) \quad (30)$$

The vector $\mathbf{e}_{\xi,i}^b$ points in the local direction along knot i , $\mathbf{e}_{\eta,i}^b$ points the other local direction in the plane, and $\mathbf{e}_z$ points in the longitudinal direction of the log and board.

Following Foley (2003), the projected fiber direction around knot i is obtained from a Rankine-oval flow pattern in the local (η_i, z_i) plane. In this representation, the knot acts as an obstacle around which the surrounding fibers are deflected. The size of the obstacle is determined from the local knot diameter $L_i(\xi_i)$, while the constants of the flow are chosen according to Foley's formulation, which is omitted here for brevity.

Let $\mathbf{u}_i^{\text{loc}}$ denote the normalized fiber direction obtained from this local flow field. This vector lies in the plane spanned by the local tangential direction and the longitudinal direction and can be written as:

$$\mathbf{u}_i^{\text{loc}} = (u_{\eta,i}, u_{z,i}) \quad (31)$$

where $u_{\eta,i}$ and $u_{z,i}$ are the corresponding components of the fiber direction. After expressing this direction in the global coordinate system, the vector becomes:

$$\mathbf{u}_i^b = u_{\eta,i} \mathbf{e}_{\eta,i}^b + u_{z,i} \mathbf{e}_z \quad (32)$$

The fiber orientation $\mathbf{u}_i^b$ does not in general lie exactly in the local growth-layer surface. Therefore, a radial component is added in the local radial direction. The full fiber direction associated with knot i is written as:

$$\mathbf{f}_i = \mathbf{u}_i^b + \lambda_i \mathbf{e}_{\xi,i}^b \quad (33)$$

where the scalar λ_i is selected so that the fiber direction satisfies the growth-layer tangency condition in equation (29). Substituting equation (33) into equation (29) gives:

$$\lambda_i = - \frac{\mathbf{u}_i^b \cdot \mathbf{n}_n}{\mathbf{e}_{\xi,i}^b \cdot \mathbf{n}_n} \quad (34)$$

This procedure gives a three-dimensional fiber direction that follows the projected flow around the knot i while remaining tangent to the surrounding growth layer.

Since the generated board may contain multiple knots, the procedure is repeated for each knot to obtain a separate three-dimensional fiber field $\mathbf{v}_i$ for each knot. These separate fields are combined by assigning a local relevance weight to each knot and then selecting the candidate that gives the strongest non-longitudinal fiber deviation. The relevance weight is based on the distance from the material point to the knot axis in the same local longitudinal-tangential plane used for the Rankine-oval flow field. This distance is defined as:

$$\ell_i = \sqrt{\Delta z_i^2 + \frac{\eta_i^2}{k_{p,i}^2}} \quad (35)$$

The distance is normalized by the local obstacle radius, taken as one half of the local knot diameter:

$$a_i = \frac{L_i(\xi_i)}{2}, \quad \rho_i = \frac{\ell_i}{a_i + \varepsilon} \quad (36)$$

The normalized distance ρ_i is therefore close to one near the boundary of the local knot obstacle and increases outside the obstacle. A smooth relevance weight is then assigned as:

$$W_i = \exp[-(\max(\rho_i - 1, 0)/\sigma)^2] \quad (37)$$

The parameter σ controls how rapidly the relevance weight decreases outside the local knot obstacle, while ε prevents division by zero where the local knot diameter becomes very small. The fiber deviation associated with the candidate $\mathbf{v}_i$ is measured by the magnitude of its transverse component in the global coordinate system:

$$D_i = \sqrt{v_{x,i}^2 + v_{y,i}^2} \quad (38)$$

A score is then formed by multiplying the local relevance weight by this transverse deviation:

$$S_i = W_i D_i \quad (39)$$

The fiber orientation selected is the one with the largest score as follows:

$$\mathbf{f}_{\log} = \mathbf{v}_{i^*}, \quad i^* = \arg \max_i S_i \quad (40)$$

Inside the body of knot i , the fiber directions $\mathbf{f}_{\text{knot},i}$ are assumed to be tangent to the knot's axis. The centerline of knot i , expressed in the global coordinate system, is written as:

$$\boldsymbol{\kappa}_i(\xi_i) = \xi_i \mathbf{e}_{\xi,i}^b + Z_{c,i}(\xi_i) \mathbf{e}_z \quad (41)$$

$$\mathbf{f}_{\text{knot},i} = \frac{\partial \boldsymbol{\kappa}_i / \partial \xi_i}{\| \partial \boldsymbol{\kappa}_i / \partial \xi_i \|} \quad (42)$$

The final 3D fiber orientation within the board is therefore given as:

$$\mathbf{f} = \begin{cases} \mathbf{f}_{\text{knot},i}, & \text{inside knot } i, \\ \mathbf{f}_{\log}, & \text{otherwise.} \end{cases} \quad (43)$$

Figure 7 shows the 3D fiber field calculated using this procedure within a board segment with multiple knots.

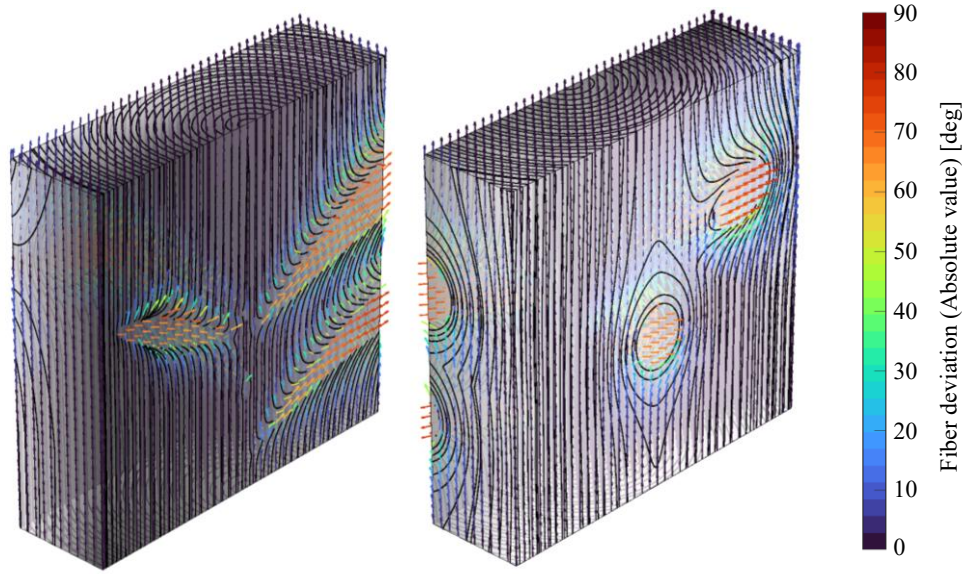


Figure 7. Three-dimensional knot field for a synthetic board segment with multiple knots.

6. Realistic timber face generator

The models presented in the previous sections generate boards with known three-dimensional knot geometry, growth layers, and fiber orientations. From this information, the growth-ring pattern on the four surfaces of a board can be obtained from the intersections between the growth layers and the planes representing the board surfaces. The 2D in-plane fiber orientation on each surface can also be obtained by projecting the 3D fiber-orientation vectors onto the corresponding surface plane.

The final step in the proposed pipeline is to convert the generated binary ring patterns and fiber maps into realistic RGB timber images. This was done using a conditional latent diffusion model trained to produce four timber faces that looks realistic while preserving the positions of annual rings, fiber deviations, knots, and knot bumps. As a compromise between visual fidelity and computational efficiency, the model was trained to produce an RGB image with a resolution of 512×512 pixels for each side of the board. The model was trained on a large number of timber boards scanned using an optical and laser scanner.

6.1 Conditional latent diffusion

Diffusion models have become one of the most widely used approaches for high-quality image generation (Rombach et al. 2021). These models are trained to reverse a gradual noising process. During training, the training images are corrupted with known amounts of random noise. The model is then trained to predict the noise or the corresponding clean image representation. During generation, the process is run in reverse: the model starts from a random noise and progressively denoises it until a realistic image is obtained.

In a latent diffusion model, the forward and reverse diffusion processes are not applied directly to the RGB pixels. Instead, a variational autoencoder (VAE) first maps the image to a lower-dimensional latent representation. The use of a latent space is important for this application because the images of the four sides are relatively large. Operating in a smaller latent space reduces the computational cost while retaining the main visual content needed to reconstruct realistic textures. At inference

time (i.e., during generation), the model starts from a random latent and repeatedly applies the learned denoising operation. The final latent is then decoded by the VAE to obtain the RGB timber face.

The objective of the present application is not simply to generate random realistic timber images, but to generate realistic images that are consistent with prescribed ring and fiber-orientation information. Therefore, a conditional latent diffusion model was used, in which additional information is provided to guide the denoising process. The model used in this work is based on a modified version of Stable Diffusion XL (SDXL) (Podell et al. 2023), which is a conditional latent diffusion model originally developed to generate high-quality realistic images from text prompts.

6.2 Stable Diffusion XL (SDXL) model

The main components of the original SDXL model are shown in Figure 8. These include a U-Net denoiser, a VAE encoder and decoder, a scheduler, a tokenizer, and a text encoder. During inference, an initial noisy latent tensor of size $4 \times 64 \times 64$ is iteratively denoised until a final clean latent tensor is obtained. This final latent tensor is then processed by the VAE decoder to generate the output RGB image of size $3 \times 512 \times 512$.

At each timestep in the iterative denoising process, the U-Net receives the current noisy latent tensor, the timestep, and conditioning information derived from the text prompt. The text prompt is processed by the tokenizer and text encoder to produce text embeddings, which are injected into the U-Net through cross-attention (Podell et al. 2023). Based on the U-Net prediction, the scheduler updates the latent tensor by taking one step toward the clean latent representation. Thus, the scheduler controls the denoising trajectory, while the U-Net provides an estimate of the noise at each step.

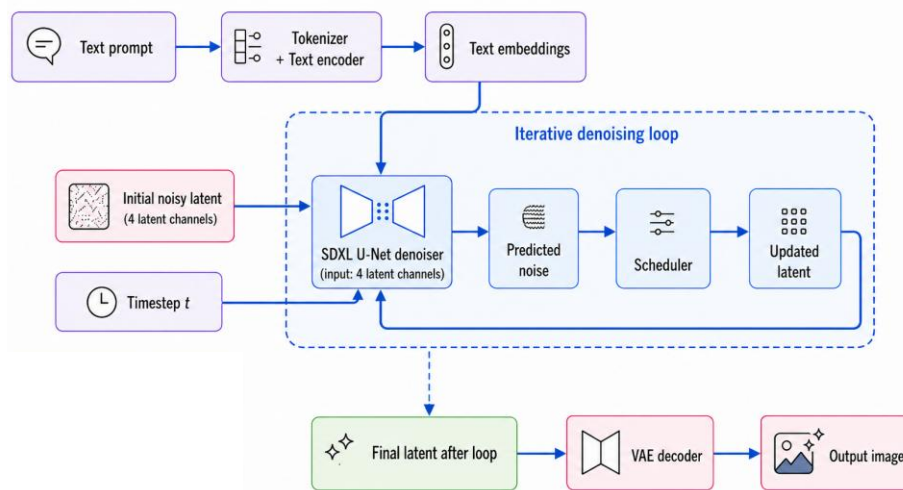


Figure 8. Schematic of the original SDXL model showing its main components.

6.3 Modified SDXL

The original SDXL model was modified to generate realistic timber images from ring and fiber information rather than from text prompts. The conditioning information consisted of four binary maps representing the surface ring patterns and four grayscale maps representing the in-plane fiber orientations on the four faces of the generated board. Figure 9 shows a schematic of the modified SDXL model.

The main modification was to expand the input of the U-Net denoiser from four latent channels to eight latent channels. The additional four channels are used to provide the conditioning information related to growth rings and fiber orientations. The 512×512 ring and fiber orientation maps are first arranged into a $4 \times 3 \times 512 \times 512$ tensor containing four face-wise condition maps. For each face s , the condition map can be written as:

$$\mathbf{C}_s = [\mathbf{R}_s, \mathbf{F}_s, \mathbf{0}_{512 \times 512}] \quad (44)$$

where $\mathbf{R}_s$ is the binary ring map, $\mathbf{F}_s$ is the grayscale map representing fiber orientations, and the third channel is a zero map. The zero map is included so that the original SDXL VAE encoder, which expects a three-channel input, could be used to encode the condition maps.

The four condition maps were processed by the SDXL VAE encoder to obtain latent condition representations (i.e., tensor of size $4 \times 4 \times 64 \times 64$). These condition latents are then concatenated with the $4 \times 4 \times 64 \times 64$ noisy image latents resulting in a tensor of size $4 \times 8 \times 64 \times 64$. The concatenated tensor is then used as initial input to the denoising loop. At each time step t , the U-Net receives a concatenation of the updated latent from the previous step and the condition latents. Since text prompts are not required in this application, null text embeddings are used in place of the text embeddings of the original SDXL model.

After the iterative denoising process, the final image latents are processed by the VAE decoder to generate four RGB images, one for each board face. In this way, the modified SDXL model produces realistic timber textures while being guided by the prescribed ring and fiber-orientation conditions.

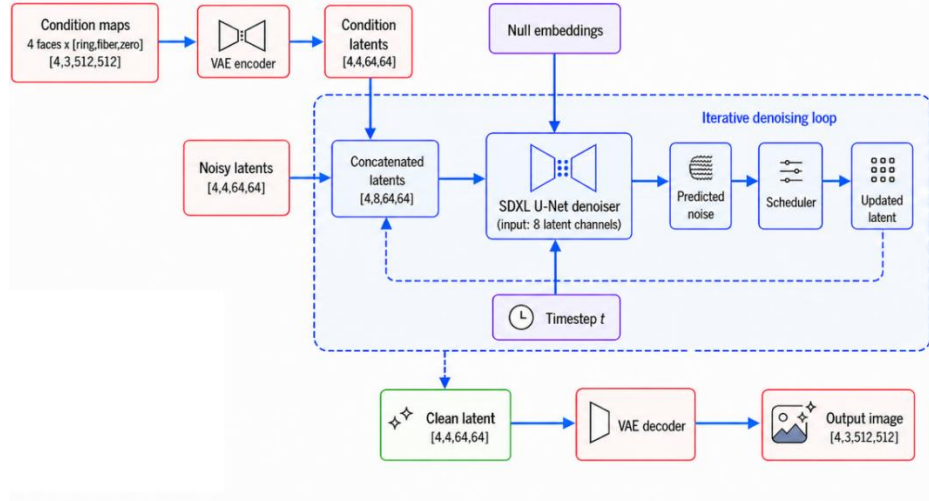


Figure 9. Schematic of the modified SDXL model.

6.4 Training dataset

The dataset used to train the modified SDXL model consisted of 1208 Norway spruce boards with nominal cross-sectional dimensions of $45 \times 145 \text{ mm}^2$. The lengths of these boards range from 3.6 to 4.8 m, with an average length of 4.31 m. All boards were scanned using an industrial wood scanner to obtain RGB images of the four longitudinal sides. The image resolution was approximately $0.8 \times 0.07 \text{ mm}^2$ in the lengthwise and crosswise directions, respectively. In addition, the scanner estimates the in-plane fiber orientation on the four sides using the tracheid effect (Kandler et al. 2016). The approximate resolution of the resulting fiber maps was $1.0 \times 4.4 \text{ mm}^2$.

The boards were segmented into 35,479 non-overlapping segments, each 145 mm long. Each segment is represented by four RGB images and four fiber orientation maps. All images and maps were resized into 512×512 pixels, which is the resolution used by the modified SDXL. The fiber maps contain angles ranging from $-\pi/2$ to $\pi/2$ radians, where zero represents a straight in-plane orientation in clear wood. These maps were rescaled to the interval [0,1] and used as grayscale images.

In addition to the RGB images and fiber maps obtained directly from the scanner, the training of the modified SDXL requires the surface ring patterns for all four faces of each segment. Therefore, the ring patterns on the four sides of 300 randomly selected board segments were manually labeled as shown in Figure 10. Manual labeling of the full dataset was not feasible because labeling annual rings requires considerable time and effort, especially for boards with tightly spaced rings.

Instead, the 300 manually labeled segments were used to train a model that automatically labels the surface ring patterns. The automatic ring-labeling model was also based on the modified SDXL architecture shown in Figure 9, but it was trained to generate four binary ring maps given four RGB images and four fiber orientation maps. This model was then used to automatically label all segments. Figure 11 shows examples of automatically labeled surfaces, illustrating good labeling performance even for boards with unusual ring patterns. The automatic labeling process made it possible to construct a much larger training dataset, allowing the model to learn from a wider range of variations in the appearance of Norway spruce surfaces. The resulting dataset consisted of 35,479 samples, each containing eight condition maps, namely the ring and fiber-orientation maps, together with the corresponding four target RGB images.

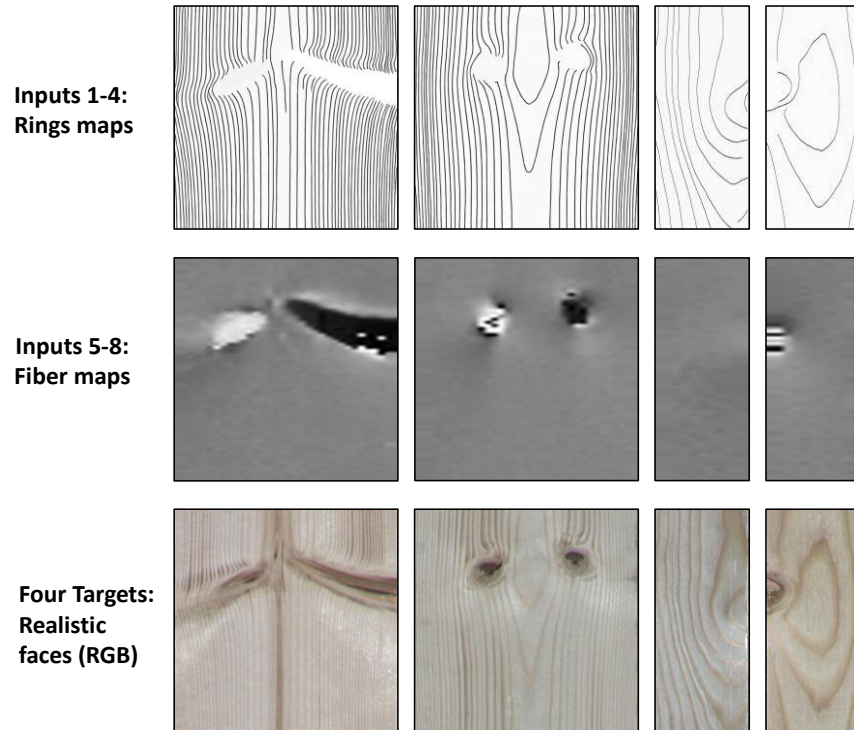


Figure 10. Inputs and targets of the training dataset used to train the realistic timber diffusion model.

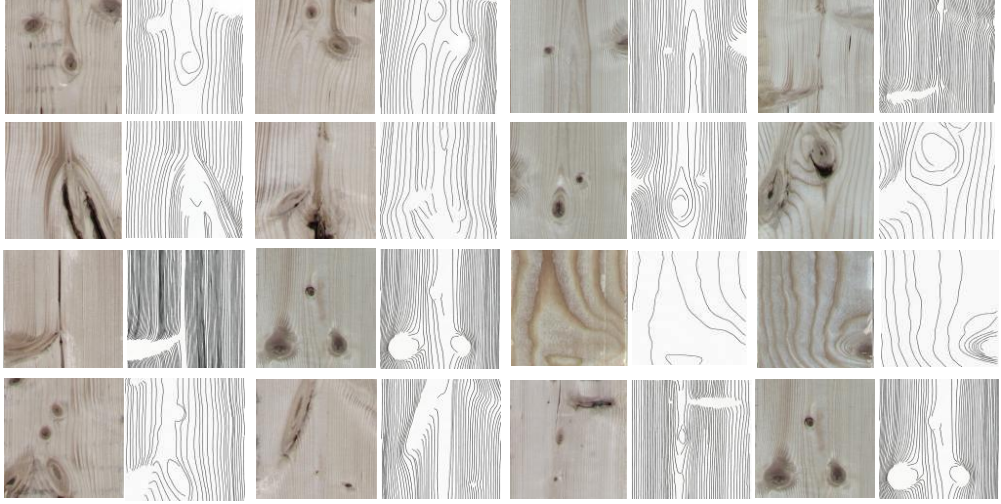


Figure 11. Examples of binary ring maps obtained using the automatic ring labeler.

6.5 Training process

During training, the $4 \times 3 \times 512 \times 512$ target maps of the training dataset are first converted into a four-channel latent representation using the SDXL VAE encoder, resulting in a $4 \times 4 \times 64 \times 64$ target latent, denoted as $\mathbf{T}_0$. The corresponding $4 \times 4 \times 64 \times 64$ condition latent, $\mathbf{T}_{\text{cond}}$ is also obtained by encoding the condition maps using the same encoder. The target latent $\mathbf{T}_0$ is then corrupted by random noise according to the following forward noising process:

$$\mathbf{T}_t = \sqrt{\bar{\alpha}_t} \mathbf{T}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon} \quad (45)$$

where $\boldsymbol{\epsilon}$ is sampled from a standard multivariate normal distribution, t is the diffusion timestep, $\bar{\alpha}_t$ is determined by the noise scheduler (a DDPM-style scheduler (Ho et al. 2020) was used in this work). The noisy latent $\mathbf{T}_t$ is then concatenated with $\mathbf{T}_{\text{cond}}$, and the denoising U-Net is trained to estimate the noise component from the $4 \times 8 \times 64 \times 64$ combined latent.

As described above, the number of input channels of the SDXL’s U-Net was expanded to eight. The weights corresponding to the original four input channels were initialized from the original SDXL UNet, while the weights of the added conditioning channels were initialized to zero. This preserves the pretrained denoising behavior at the beginning of training while allowing the model to gradually learn how to use the ring and fiber conditions.

The U-Net was trained by minimizing a loss function composed of a noise-prediction term and a latent-reconstruction term:

$$L = L_{\text{noise}} + \lambda_{\text{rec}} L_{\text{rec}} \quad (46)$$

where λ_{rec} controls the relative contribution of the reconstruction loss. The noise-prediction loss encourages the noise predicted by the U-Net to match the true noise added to the target latent, while the reconstruction loss ensures that the final clean latent remains close to the target latent.

The noise-prediction loss is calculated as the expected weighted squared error between the noise predicted by the U-Net and the true noise:

$$L_{\text{noise}} = \mathbb{E}_{t, \boldsymbol{\epsilon}} [w(t) \|\boldsymbol{\epsilon}_{\theta}([\mathbf{T}_t, \mathbf{T}_{\text{cond}}], t) - \boldsymbol{\epsilon}\|_2^2] \quad (47)$$

where $\epsilon_\theta([T_t, T_{\text{cond}}], t)$ is the noise predicted by the U-Net at the timestep t using the concatenated latent $[T_t, T_{\text{cond}}]$, and $w(t)$ is the timestep-dependent weight. In this work, min-SNR weighting (Hang et al. 2023) was adopted.

The reconstruction loss L_{rec} is calculated by firstly reconstructing an estimate of the clean latent, $\hat{T}_0$, from the noisy latent and the predicted noise ϵ_θ as follows:

$$\hat{T}_0 = \frac{T_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta}{\sqrt{\bar{\alpha}_t}} \quad (48)$$

This estimate is then used to compute L_{rec} , which uses smooth L1 loss to calculate the error between the target and reconstructed latents:

$$L_{\text{rec}} = \text{SmoothL1}(\hat{T}_0, T_0) \quad (49)$$

Classifier-free guidance (Ho and Salimans 2022) was used to control how strongly the generated images follow the ring and fiber condition maps. During training, the condition latent T_{cond} was randomly dropped for a 10% of the training examples. The purpose was to teach the U-Net to denoise both with and without conditioning information. During inference, the model is evaluated twice at each timestep: once with the condition latent and once without it. The two predicted noise components are then combined as:

$$\hat{\epsilon} = \epsilon_{\text{uncond}} + g_s(\epsilon_{\text{cond}} - \epsilon_{\text{uncond}}) \quad (50)$$

$$\epsilon_{\text{cond}} = \epsilon_\theta([T_t, T_{\text{cond}}], t), \quad \epsilon_{\text{uncond}} = \epsilon_\theta([T_t, 0], t) \quad (51)$$

where g_s is the guidance scale. Larger values of g_s enforce stronger agreement with the prescribed ring and fiber-orientation maps, whereas smaller values allow more freedom in the generated color and texture.

The modified SDXL model was trained in PyTorch (Paszke et al. 2019) using two NVIDIA RTX A6000 GPUs. The dataset was split into training and validation subsets using a 90/10 split, resulting in 31,930 training samples and 3,548 validation samples. The optimizer was Adafactor with a learning rate of 2×10^{-5} , a cosine learning-rate schedule, and 2000 warm-up steps. The effective batch size was four board segments. The model was trained for 60,000 optimization steps with 1000 noising timesteps, and the weight assigned to the latent-reconstruction loss was $\lambda_{\text{rec}} = 0.1$.

7. Preliminary application of the board generator

The proposed board generator is expected to have considerable potential as a source of synthetic data for training data-driven models for wood assessment and grading. As discussed in the introduction, many tasks of practical interest require information that is difficult, expensive, or even impossible to obtain at scale from real timber boards. In such cases, synthetic boards with known internal structure, including knots, growth layers, fiber orientations, and surface appearance can be used to create paired datasets where both the external scanner information and the corresponding internal or hidden target quantities are known.

This section presents one preliminary application to demonstrate the practical potential of the proposed board generator. The selected application is the prediction of the cross-sectional appearance of a timber board at an arbitrary longitudinal

position using only images of the four longitudinal sides obtained by optical scanning. A model was trained exclusively on synthetic board segments generated by the proposed board generator and was then evaluated on real Norway spruce boards.

This application is practically relevant because the cross-section contains information about the pith location and the annual ring pattern. Both quantities are important for the assessment and modeling of timber. In softwood species, clear-wood density, longitudinal modulus of elasticity, and modulus of rupture generally increase from pith to bark (Abdeljaber et al. 2023). Pith location also determines the radial orientation of knots, which affects local fiber orientation and therefore the stiffness and strength of sawn timber (Kandler et al. 2016). In addition, it influences shape stability because moisture-related distortions such as cupping, twisting, and bowing depend on local material directions, which are defined relative to the pith (Ormarsson et al. 1999). The pith position can also affect visual quality, since boards with visible or enclosed pith may receive lower grades or be rejected in certain applications (EN 1611–1:1999 1999). Annual ring widths are also related to several material properties in softwood, including density, stiffness, and strength. Wider annual rings are generally associated with lower density and stiffness in clear wood, and several visual grading standards for softwood impose limits on average annual ring width (Abdeljaber et al. 2023).

7.1 Synthetic training dataset

The board generator was used to generate 30,000 synthetic Norway spruce board segments with randomized knots, annual rings, crook, taper, and position with respect to the pith. The dimension of the generated segments was $45 \times 145 \times 145 \text{ mm}^3$. The diffusion model was used to apply realistic spruce faces on the generated boards based on their ring patterns and in-plane fiber orientations. Examples of the generated boards are shown in Figure 12. The guidance scale g_s , was randomly picked within the interval $[1.00, 2.50]$. The purpose was to generate wood texture with varying degrees of randomness. As explained in Section 6.5, large guidance values force the generated surfaces to follow the prescribed condition maps more closely, whereas smaller values allow more freedom in color and texture generation. The effect of g_s is illustrated in Figure 13.

A training dataset containing 30,000 input-target pairs was obtained from the generated board segments. Each input consists of four RGB images corresponding to the four sides of a segment and each target is a single binary image representing the bottom cross-section of the segment. The input RGB images were rotated and oriented toward the cross-section as shown in Figure 14 to make it easier for the cross-section prediction model to learn the relationship between the face ring pattern and the shape of the annual ring at the cross-section's edge. All input and target images were then resized into 512×512 .



Figure 12. Examples of synthetic board segments generated using the proposed framework.

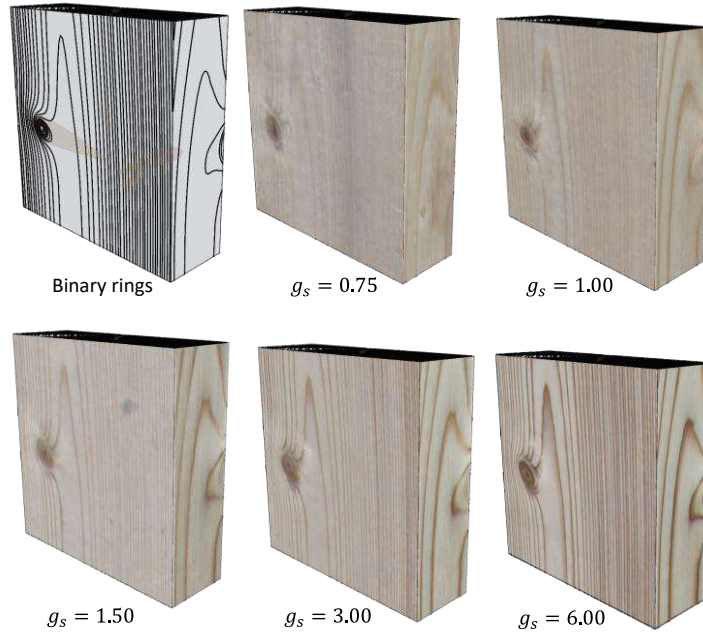


Figure 13. Example showing the effect of the guidance scale on the generated wood texture.

7.2 Testing dataset from real boards

The testing set was obtained from 76 real Norway spruce boards that were not included in the training of the diffusion model used to produce the synthetic boards. The cross-sections at both ends of each board were scanned using a flatbed scanner, resulting in 152 cross-section images. The nominal cross-sectional dimension of the testing boards was $45 \times 145 \text{ mm}^2$. Examples of the scanned cross-sections are shown in Figures 14 and 15. Further details about the boards used for testing can be found in (Abdeljaber et al. 2023).

7.3 Training of the cross-section prediction model

Our first attempt to train a model that predicts the cross-sectional appearance from the four board surfaces was to train a conventional U-Net model, consisting of an

encoder and a decoder. The U-Net encoder used three downsampling stages with channel widths 64, 128, and 256, followed by a 256-channel bottleneck. Each convolution block consisted of two 3×3 convolutions with GroupNorm and SiLU activation. The decoder used bilinear upsampling, 1×1 convolutions, skip connections from the matching encoder levels, and convolution blocks with widths 256, 128, and 64 before a final 1×1 convolution produced one output logit map of size $1\times 512\times 512$, which was converted to a bottom-ring probability map by a sigmoid function. The model has 4.76 million trainable parameters.

The dataset was split into training and validation subsets using a 90/10 split, resulting in 27,000 training samples and 3,000 validation samples. The loss was taken as the binary cross-entropy between the output and target maps. The training was conducted on PyTorch (Paszke et al. 2019) using the same hardware described in Section 6.5. The validation loss was monitored after each epoch, and training was stopped after 79 epochs when no further improvement was observed. The model was then evaluated on both synthetic validation samples and the real board segments described in Section 7.2.

Figures 14 and 15 shows examples of the U-Net predictions. For both synthetic and real inputs, the U-Net produced sharp and accurate predictions near the cross-section edges. However, the inner portions of the predicted cross-sections were blurry. This can be explained by the fact that the cross-section is strongly constrained near the edges, where the annual rings should coincide with the ring patterns on the adjacent faces. The inner portion, on the other hand, is less constrained because many different internal ring patterns may be compatible with the observed surface information. The U-Net tends to average these possible cross-sectional appearances, which results in blurry output.

The second attempt was to train a conditional latent diffusion model similar to the modified SDXL described in Section 6.3, but with 12 input channels corresponding to the four RGB images and a single output channel representing the cross-sectional rings. The model was trained on the same synthetic dataset. The training loss and hyperparameters were similar to those described in Section 6.5. After training, the model was evaluated on both synthetic and real board segments. As shown in Figure 14, the model was able to generate non-blurry predictions. However, it failed to produce structured rings and instead hallucinated highly irregular patterns that do not resemble actual wood cross-sections.

To address the limitations observed in both the U-Net and diffusion predictors, a combined pipeline was developed. In this approach, the U-Net predictor was first used to generate an initial cross-section image, and a diffusion model was then trained to refine that prediction. The idea was to use the U-Net prediction as an additional condition, giving the diffusion model a more constrained starting point and reducing the risk of hallucinated ring patterns. The U-Net used in this pipeline was the same model described above. The diffusion refiner was a modified SDXL model with 13 input channels, where the additional channel was used to provide the U-Net prediction as an extra condition. The refiner model was trained on the same synthetic dataset, with the additional condition map obtained from the U-Net predictor. The results of the combined cross-section prediction approach are shown in Figure 15. The predicted cross-sections from real boards appear visually sharp and consistent with the actual cross-sections, even within the inner portions.

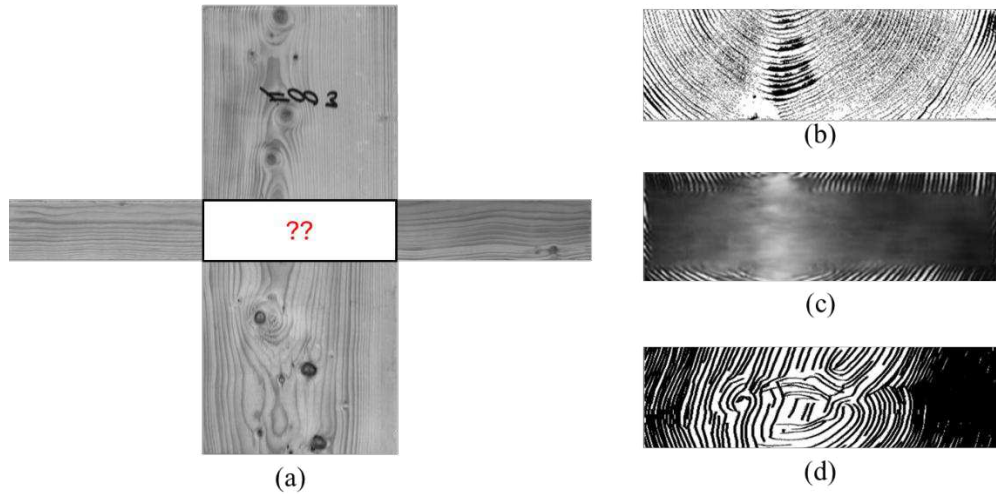


Figure 14. Example of poor annual-ring prediction. (a) Input to the cross-section predictor, consisting of four board-face images oriented toward the target cross-section. (b) Ground-truth cross-section. (c) Cross-section predicted by the plain U-Net model. (d) Cross-section generated by the plain diffusion model, showing a highly irregular hallucinated annual ring pattern.

7.4 Evaluation of the predicted cross-sections

To evaluate the accuracy of the cross-sections predicted by the combined pipeline, the method was applied to the 152 end segments of the testing boards. The annual rings in both the actual cross-section images and the predicted images were manually labeled using the approach developed by Abdeljaber et al. (2023). The pith location and average annual ring width were then calculated for each cross-section. Table 2 and Figure 16 present a statistical analysis of the discrepancies between the values obtained from the actual and predicted cross-sections.

Overall, the results show that the predicted ring widths are well aligned with those measured from the actual cross-sections, with a coefficient of determination R^2 of 0.93 and a mean absolute discrepancy of 0.173 mm. The pith locations estimated from the predicted cross-sections are also in good agreement with the corresponding values estimated from the actual cross-sections, with mean absolute discrepancies of 3.77 mm in the x -direction and 6.08 mm in the y -direction.

Table 2. Statistical analysis of the absolute discrepancies between measurements from the actual cross-sections and the corresponding values derived from the predicted cross-sections. Reported quantities include the average annual-ring width and the pith coordinates in the x - and y -directions.

Quantity	Average ring width [mm]	Pith- x [mm]	Pith- y [mm]
Mean	0.173	3.77	6.08
Median	0.130	2.43	4.12
Std.	0.170	4.60	6.02
85 th percentile	0.295	6.13	10.0
90 th percentile	0.377	7.32	13.5
95 th percentile	0.574	12.7	19.2

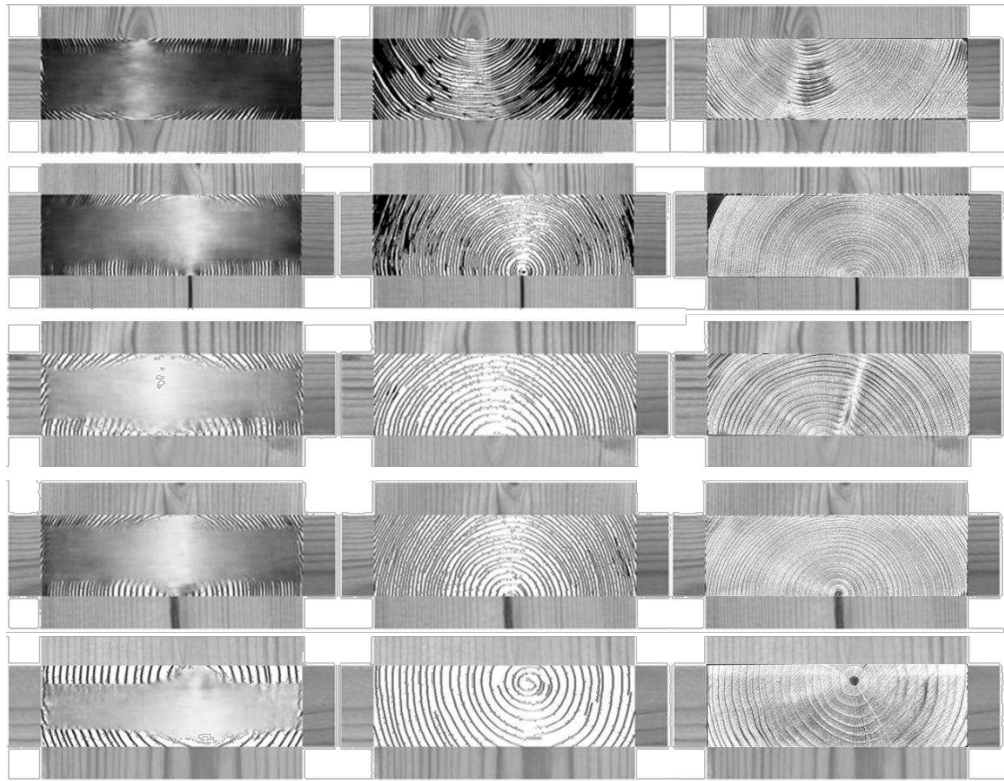


Figure 15. Comparison of predicted and actual cross-sections. The first column shows examples of blurry cross-section images predicted by the plain U-Net model. The second column shows the corresponding refined predictions produced by the combined pipeline, in which the U-Net output is further processed by the SDXL refiner. The third column shows the actual cross-sections.

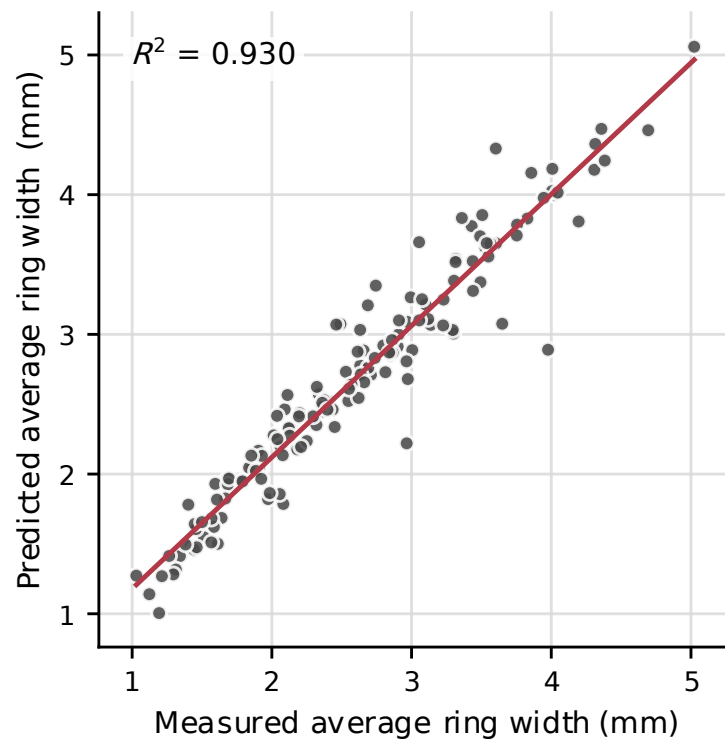


Figure 16. Comparison between measured and predicted average annual ring widths. Each point represents one cross-section, and the red line indicates the linear fit.

8. Discussion and limitations

The proposed board generator is arguably the first complete framework for generating synthetic timber boards with known knot geometry, growth layers, fiber orientations, and realistic external surfaces. The framework combines four components, including a knot sequence generator, a growth-layer model, a fiber-orientation model, and a conditional diffusion model for surface appearance. Each component contributes to the overall realism of the generated boards, but each also has limitations that could be addressed in more focused future studies.

The knot generator was trained on short overlapping sequences extracted from longer sequences representing full logs. In the present formulation, the longitudinal position of a knot was not included as a component of the knot embedding vector. Instead, the longitudinal position was determined by the position of the knot token within the generated sequence. This implies that the geometric properties of a knot were assumed to depend on the local sequence context rather than on global longitudinal position of the knot in the tree. In real trees, however, knot properties such as size, curvature, and radial extent may be correlated with the height. A larger dataset of long knot sequences would make it possible to develop an enhanced knot generator in which longitudinal position is included explicitly, allowing for capturing possible correlations between knot geometry and the position within the log.

The growth-layer generator also involves some simplifying assumptions. In particular, the distributions used to sample crook and taper parameters were selected manually rather than derived directly from measurements of real logs. In future work, these distributions could be estimated from larger datasets of scanned logs. This limitation is not expected to be critical since most data-driven applications require only short board segments rather than full-length logs. Over such short segments, the effects of crook and taper are limited. Nevertheless, data-driven calibration of crook and taper would be important for applications that require accurate full-length board or log geometry.

The first three components of the board generator, namely the knot, growth-layer, and fiber-orientation models, can in principle be used to generate logs and boards of arbitrary dimensions. However, the diffusion model used to generate realistic timber faces was only trained for the specific dimensions of $145 \times 145 \text{ mm}^2$ for wide faces and $45 \times 145 \text{ mm}^2$ for narrow faces. Adapting the surface-generation model to other board dimensions, or aspect ratios would require training data from boards with more varied nominal dimensions. Future work could investigate models that are less tied to a fixed image size or aspect ratio.

The image resolution used in the present work, 512×512 pixels per face, was sufficient for generating realistic surface texture, growth-ring patterns, and overall knot appearance. However, some fine details, such as tight ring patterns and small cracks inside knots may require higher resolution training images. Training diffusion models at higher resolution is computationally demanding, especially when multiple board faces are generated simultaneously. For the intended use of the generator as a source of training data, the present resolution may already be sufficient for many applications.

A further limitation is that the entire framework was developed using data and images from Norway spruce. Transferring the generator to other species would require species-specific adaptation. For closely related species, particularly other spruce

species, the changes may be relatively limited and may mainly involve fine-tuning the surface-generation model using scanned boards from the target species. For related conifers such as fir or pine, additional adjustments may be needed because knot appearance, heartwood-sapwood contrast and ring patterns can differ more noticeably. A practical approach for such adaptations could be to train a lightweight adapter, such as a low-rank adaptation (LoRA) module (Hu et al. 2021), on top of the existing diffusion model. This would allow the surface-generation model to learn species-specific color and texture characteristics without retraining the full model. The required dataset for such fine-tuning would likely be much smaller than the dataset used for the original model. On the other hand, transferring the full generator to species with substantially different anatomy or surface appearance, such as hardwoods, would likely require species-specific data, retraining of the surface model, and modification of the analytical models for knots, growth layers, and fiber orientation.

The preliminary application presented in Section 7 showed that a model trained only on generated board segments could predict cross-sectional ring patterns of real Norway spruce boards with promising accuracy. This supports the idea that generated boards can be useful for training data-driven models for tasks where paired real data are difficult to collect. However, this result should not be interpreted as evidence that synthetic training data will be sufficient for all applications. For more complex tasks, such as predicting three-dimensional knot geometry or full-volume three-dimensional fiber orientation from external scanner data, the domain gap between generated and real boards may be more substantial. In such cases, models trained on generated data may need to be fine-tuned using examples from real boards. Even then, the generated data can reduce the amount of real paired data required and can provide a useful basis for feasibility studies before committing to expensive data collection campaigns.

The cross-section prediction application itself also has several limitations. First, the evaluation was carried out on a relatively limited set of real boards, consisting of 152 end cross-sections from 76 boards. Second, the combined U-Net and diffusion-refiner pipeline is computationally more expensive than a purely feed-forward model, because the diffusion refiner requires repeated denoising iterations. This makes the current implementation not directly suitable for real-time sawmill operation. Future work could investigate faster diffusion samplers, or purely feed-forward models. Third, the output of the model is a predicted cross-section image. To use this information in grading or process control, additional postprocessing is required to extract quantities such as pith location and annual ring width. For these reasons, the application in Section 7 should be viewed as a proof of concept rather than a final industrial solution. Its purpose was to demonstrate that the proposed generator can provide useful training data for a task where paired real data would be difficult to obtain at scale.

9. Conclusion

This study presented a framework for procedural generation of synthetic Norway spruce boards with realistic surface appearance and known internal structure. The framework combines a data-driven knot sequence generator, a growth-layer model, a fiber-orientation model, and a conditional diffusion model for generating realistic timber surfaces from prescribed ring and fiber-orientation maps. Together, these

components make it possible to generate synthetic boards with known knot geometry, growth-layer geometry, fiber orientations, and scanner-like RGB surface images.

The generated boards were also used in a preliminary application, where a model was trained to predict timber cross-sections from images of the four external board surfaces. The model was trained exclusively on synthetic board segments and evaluated on real Norway spruce boards. The results indicate that generated boards can provide useful paired training data for applications where real ground truth would be difficult, costly, or destructive to obtain.

It is important to note that the degree to which generated boards are useful for training DL models will depend on the complexity of the target application. In some cases, such as the preliminary application presented in this study, models trained only on generated boards may be sufficient. In other cases, generated boards may be more suitable as a pretraining or feasibility tool, reducing the amount of real paired data needed before fine-tuning and validation on real boards. The proposed generator is therefore not intended to fully replace experimental data, but rather to support the development and testing of learning-based methods when it is difficult to obtain real data.

The board generator, including the source code and trained models, has been made open-source and is available at: https://github.com/osamaoa/board_generator. The repository includes a graphical user interface for generating and inspecting individual boards, as well as a command-line interface for automatically generating large datasets of synthetic boards.

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References

- Abdeljaber O, Habite T, Olsson A (2023) Automatic estimation of annual ring profiles in Norway spruce timber boards using optical scanning and deep learning. *Comput Struct* 275:106912. <https://doi.org/10.1016/j.compstruc.2022.106912>
- Besseau B, Pot G, Collet R, Viguié J (2020) Influence of wood anatomy on fiber orientation measurement obtained by laser scanning on five European species. *Journal of Wood Science* 2020 66:1 66:74-. <https://doi.org/10.1186/S10086-020-01922-Y>
- Blouin D, Beaulieu J, Daoust G, Poliquin J (1994) Wood quality of Norway spruce grown in plantations in Quebec. *Wood and Fiber Science* 26:342–353
- Briggert A, Olsson A, Oscarsson J (2020) Prediction of tensile strength of sawn timber: definitions and performance of indicating properties based on surface laser scanning and dynamic excitation. *Materials and Structures/Materiaux et Constructions* 53:. <https://doi.org/10.1617/s11527-020-01460-5>
- Ding F, Zhuang Z, Liu Y, et al (2020) Detecting Defects on Solid Wood Panels Based on an Improved SSD Algorithm. *Sensors* 2020, Vol 20, Page 5315 20:5315. <https://doi.org/10.3390/S20185315>
- Dinwoodie JM (2002) *Timber: Its nature and behaviour*, Second edition. Timber: Its nature and behaviour, Second edition 1–257. <https://doi.org/10.4324/9780203477878>

- EN 1611–1:1999 (1999) Sawn timber—appearance grading of softwoods Part 1: European spruces, firs, pines and Douglas firs.
- Fang Y, Guo X, Chen K, et al (2021) Accurate and automated detection of surface knots on sawn timbers using YOLO-V5 model. *BioResources* 16:5390–5406
- Foley C (2003) Modeling the effects of knots in Structural Timber. Lund University
- Frid-Adar M, Diamant I, Klang E, et al (2018) GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification. *Neurocomputing* 321:321–331. <https://doi.org/10.1016/J.NEUCOM.2018.09.013>
- Habite T, Abdeljaber O, Olsson A (2022) Determination of pith location along Norway spruce timber boards using one dimensional convolutional neural networks trained on virtual timber boards. *Constr Build Mater* 329:127129. <https://doi.org/10.1016/J.CONBUILDMAT.2022.127129>
- Habite T, Olsson A, Oscarsson J (2020) Automatic detection of pith location along Norway spruce timber boards on the basis of optical scanning. *European Journal of Wood and Wood Products* 78:1061–1074. <https://doi.org/10.1007/s00107-020-01558-1>
- Hang T, Gu S, Li C, et al (2023) Efficient Diffusion Training via Min-SNR Weighting Strategy. 7441–7451
- Ho J, Jain A, Abbeel P (2020) Denoising Diffusion Probabilistic Models. *Adv Neural Inf Process Syst* 2020-December:
- Ho J, Salimans T (2022) Classifier-Free Diffusion Guidance
- Hu EJ, Shen Y, Wallis P, et al (2021) LoRA: Low-Rank Adaptation of Large Language Models
- Hu M, Olsson A, Abdeljaber O, Huber J (2025) Determining growth surfaces and fiber orientation in Norway spruce sawn timber using X-ray computed tomography and optical scanning. *Constr Build Mater* 482:141734. <https://doi.org/https://doi.org/10.1016/j.conbuildmat.2025.141734>
- Huber JAJ, Broman O, Ekevad M, et al (2022) A method for generating finite element models of wood boards from X-ray computed tomography scans. *Comput Struct* 260:. <https://doi.org/10.1016/j.compstruc.2021.106702>
- Isaksson T (1999) Modelling the Variability of Bending Strength in Structural Timber - Length and Load Configuration Effects. Lund University
- Kandler G, Lukacevic M, Füssl J (2016) An algorithm for the geometric reconstruction of knots within timber boards based on fibre angle measurements. *Constr Build Mater* 124:945–960. <https://doi.org/10.1016/J.CONBUILDMAT.2016.08.001>
- Larsson M, Ijiri T, Yoshida H, et al (2022) Procedural texturing of solid wood with knots. *ACM Trans Graph* 41:. <https://doi.org/10.1145/3528223.3530081>
- Longuetaud F, Leban JM, Mothe F, et al (2004) Automatic detection of pith on CT images of spruce logs. *Comput Electron Agric* 44:107–119. <https://doi.org/10.1016/j.compag.2004.03.005>
- Longuetaud F, Schraml R, Mothe F, et al (2023) Traceability and quality assessment of Norway spruce (*Picea abies* (L.) H.Karst.) logs: the TreeTrace_spruce database. *Ann For Sci* 80:9. <https://doi.org/10.1186/s13595-023-01178-8>
- Ormarsson S, Dahlblom O, Petersson H (2000) A numerical study of the shape stability of sawn timber subjected to moisture variation. *Wood Sci Technol* 34:207–219. <https://doi.org/10.1007/s002260000042>
- Ormarsson S, Dahlblom O, Petersson H (1999) A numerical study of the shape stability of sawn timber subjected to moisture variation Part 2: Simulation of drying board. *Wood Sci Technol* 33:407–423. <https://doi.org/10.1007/s002260050126>
- Paszke A, Gross S, Massa F, et al (2019) PyTorch: An Imperative Style, High-Performance Deep Learning Library. *Adv Neural Inf Process Syst* 32:

- Penvern H, Demoulin L, Pot G, et al (2024) A laboratory method to determine 3D fibre orientation around knots in sawn timber: case study on a Douglas fir specimen. *Wood Sci Technol* 58:1735–1760. <https://doi.org/10.1007/s00226-024-01583-w>
- Podell D, English Z, Lacey K, et al (2023) SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis. 12th International Conference on Learning Representations, ICLR 2024
- Richter SR, Vineet V, Roth S, Koltun V (2016) Playing for Data: Ground Truth from Computer Games. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 9906 LNCS:102–118. https://doi.org/10.1007/978-3-319-46475-6_7
- Rombach R, Blattmann A, Lorenz D, et al (2021) High-Resolution Image Synthesis with Latent Diffusion Models. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition 2022-June*:10674–10685. <https://doi.org/10.1109/CVPR52688.2022.01042>
- Ros G, Sellart L, Materzynska J, et al (2016) The SYNTHIA Dataset: A Large Collection of Synthetic Images for Semantic Segmentation of Urban Scenes. 3234–3243
- Sharma RP, Vacek Z, Vacek S (2016) Modeling individual tree height to diameter ratio for Norway spruce and European beech in Czech Republic. *Trees* 2016 30:6 30:1969–1982. <https://doi.org/10.1007/S00468-016-1425-2>
- Urbonas A, Raudonis V, Maskeliunas R, Damaševičius R (2019) Automated Identification of Wood Veneer Surface Defects Using Faster Region-Based Convolutional Neural Network with Data Augmentation and Transfer Learning. *Applied Sciences* 2019, Vol 9, Page 4898 9:4898. <https://doi.org/10.3390/APP9224898>
- Zhou Z, Wellinghoff J, Fan M, et al (2025) CTPro: CT Image Quality Evaluation and Protocol Optimization
- Zhu Y, Xu Z, Lin Y, et al (2024) A Multi-Source Data Fusion Network for Wood Surface Broken Defect Segmentation. *Sensors* 2024, Vol 24, Page 1635 24:1635. <https://doi.org/10.3390/S24051635>